

4: Normal distributions, correlations

Frank Edwards

Random Variables

Random Variables

- Random variables assign values to events
- Each value is mutually exclusive
- The set of all values is exhaustive (the sample space Ω)
- Discrete random variables take a finite number of values (e.g. TRUE, FALSE)
- Continuous random variables are real numbers, and take on an infinite number of values

The expectation of a random variable

- The expectation of a random variable $E(X)$ is the mean.

Variance and standard deviation

- The standard deviation sd measures the average difference of observations from the mean

$$sd = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

- Sometimes we use the variance, which is sd^2

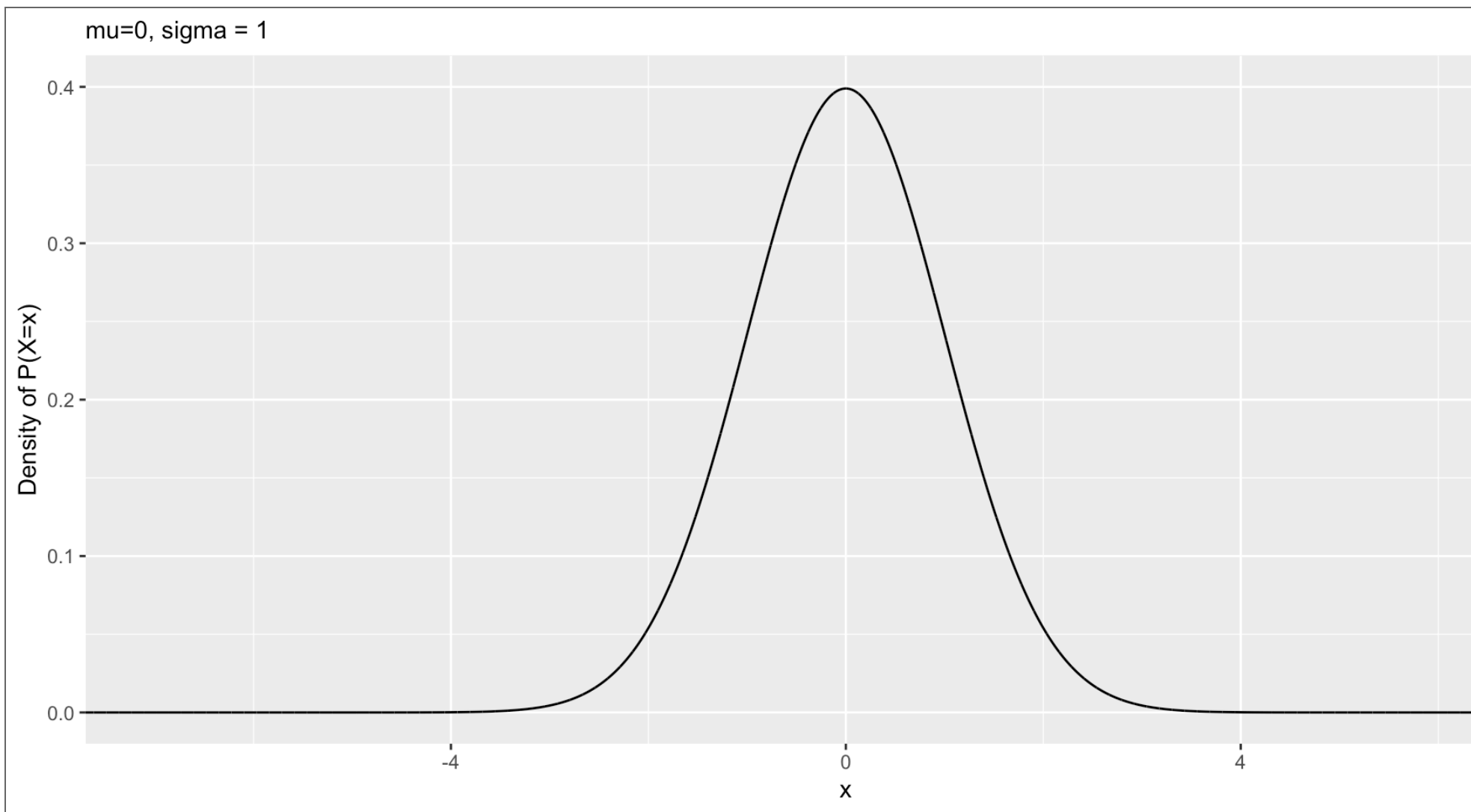
The Normal Distribution

The Normal (Gaussian) distribution is continuous, and takes on values from $[-\infty, \infty]$. It has two parameters, the mean μ and standard deviation σ (or variance σ^2).

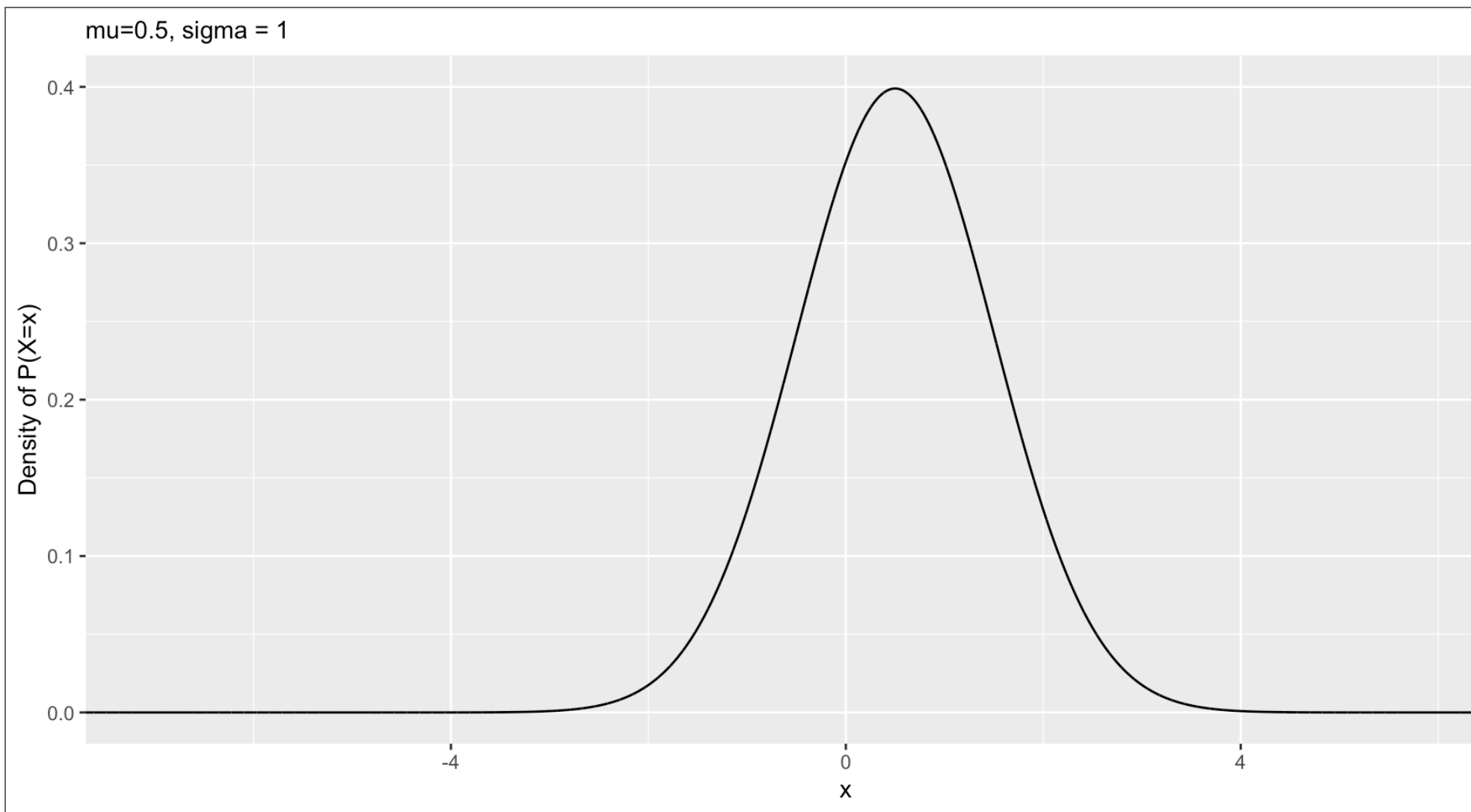
- μ determines the location of the distribution
- σ determines the spread of the distribution

The location parameter

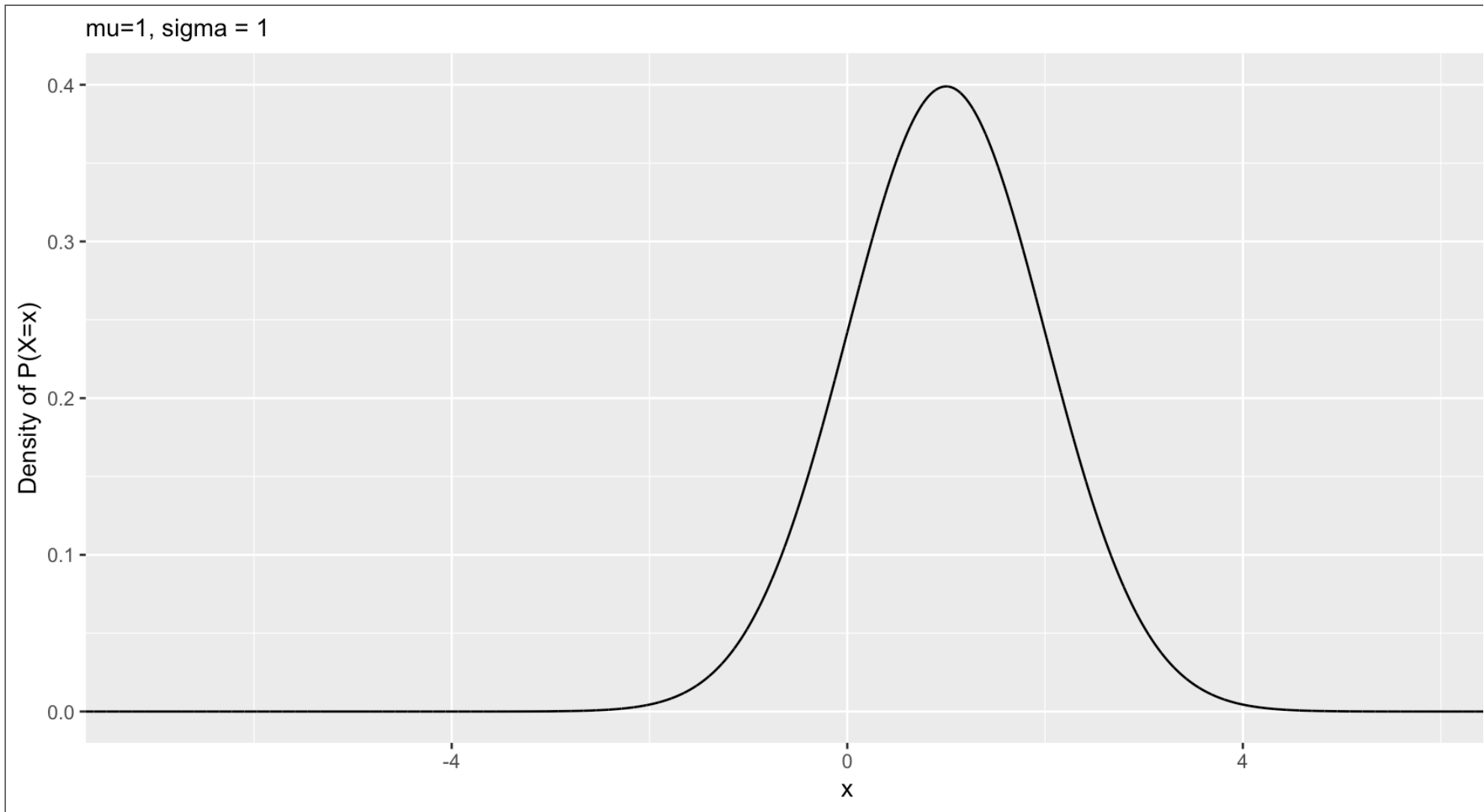
The Normal PDF



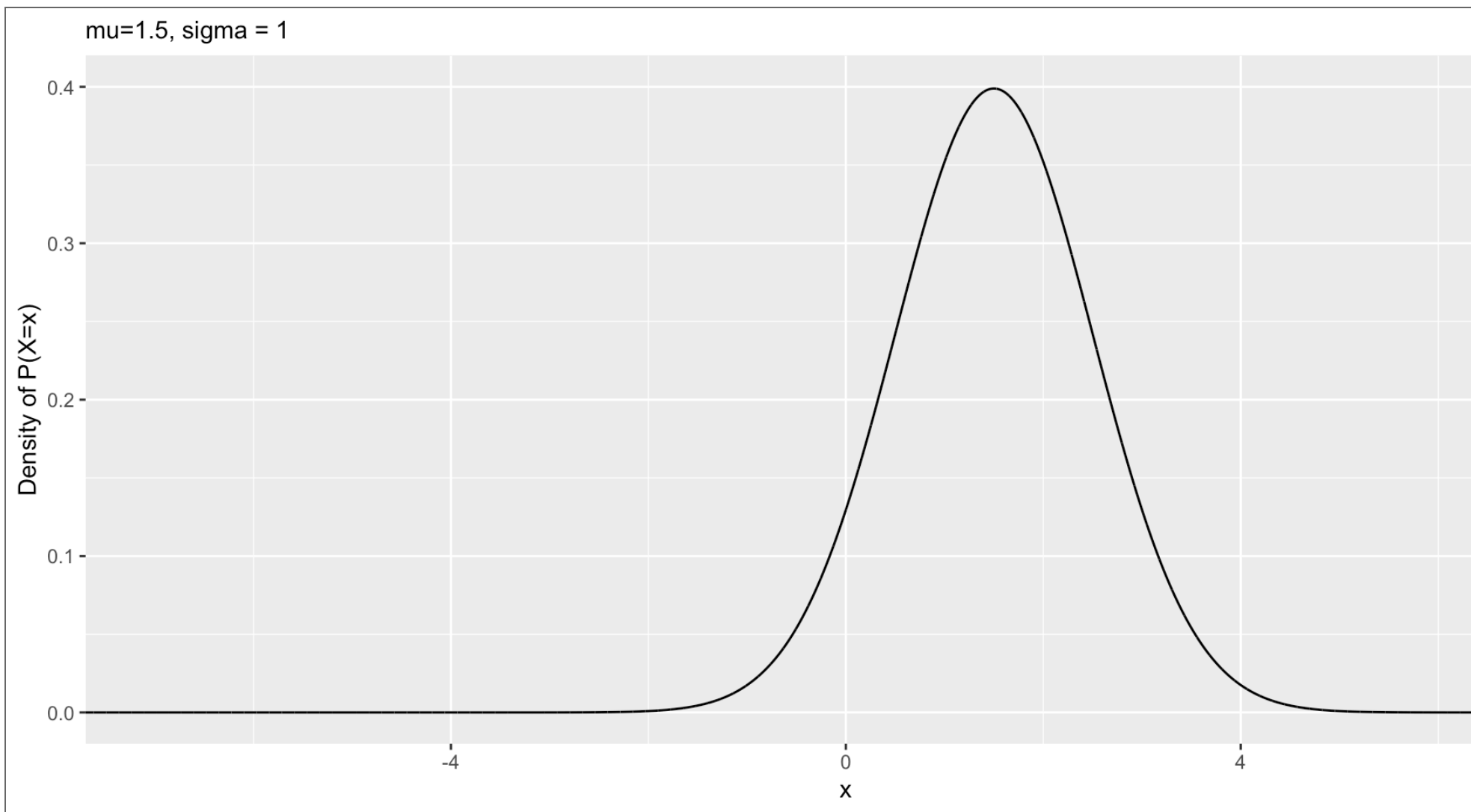
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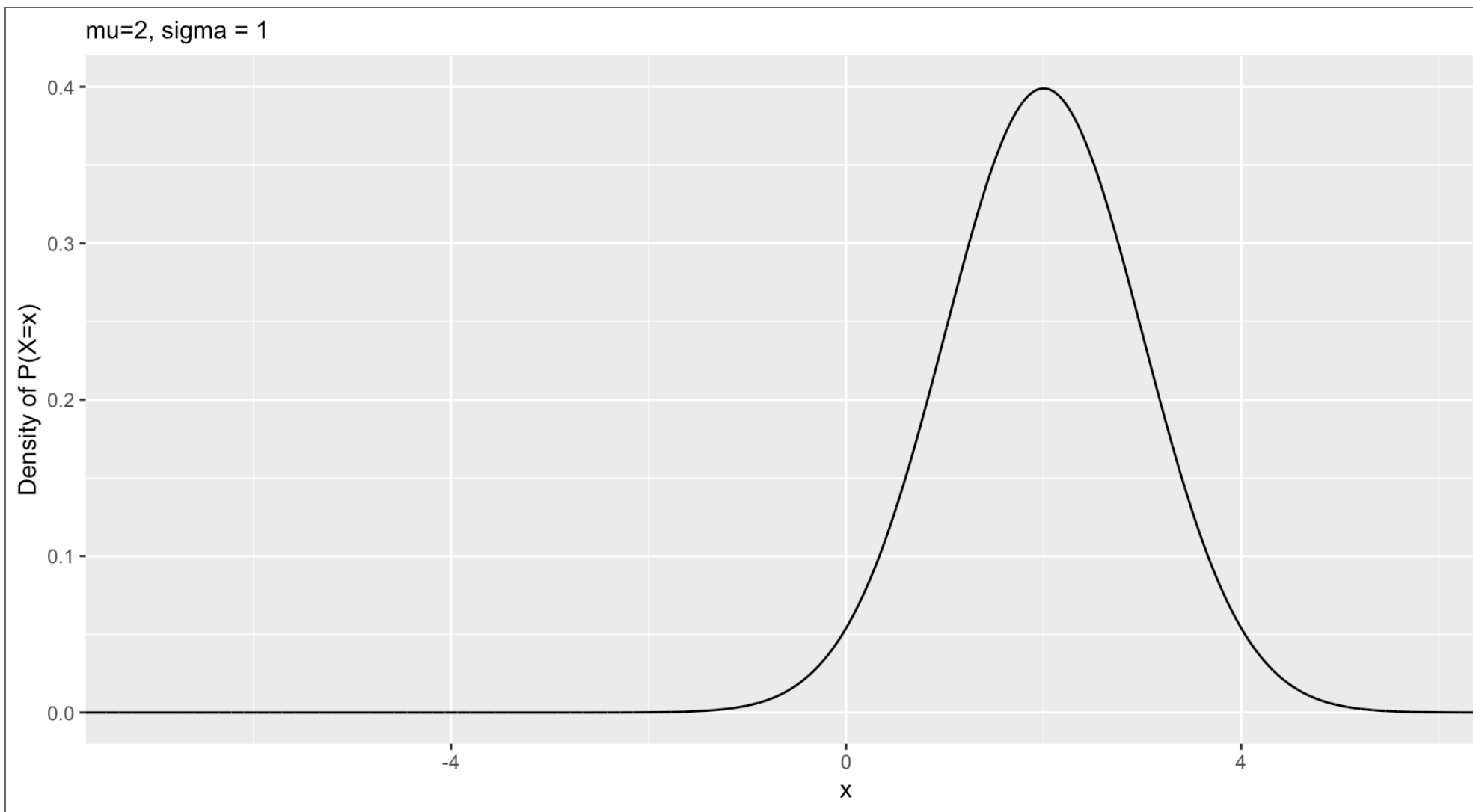
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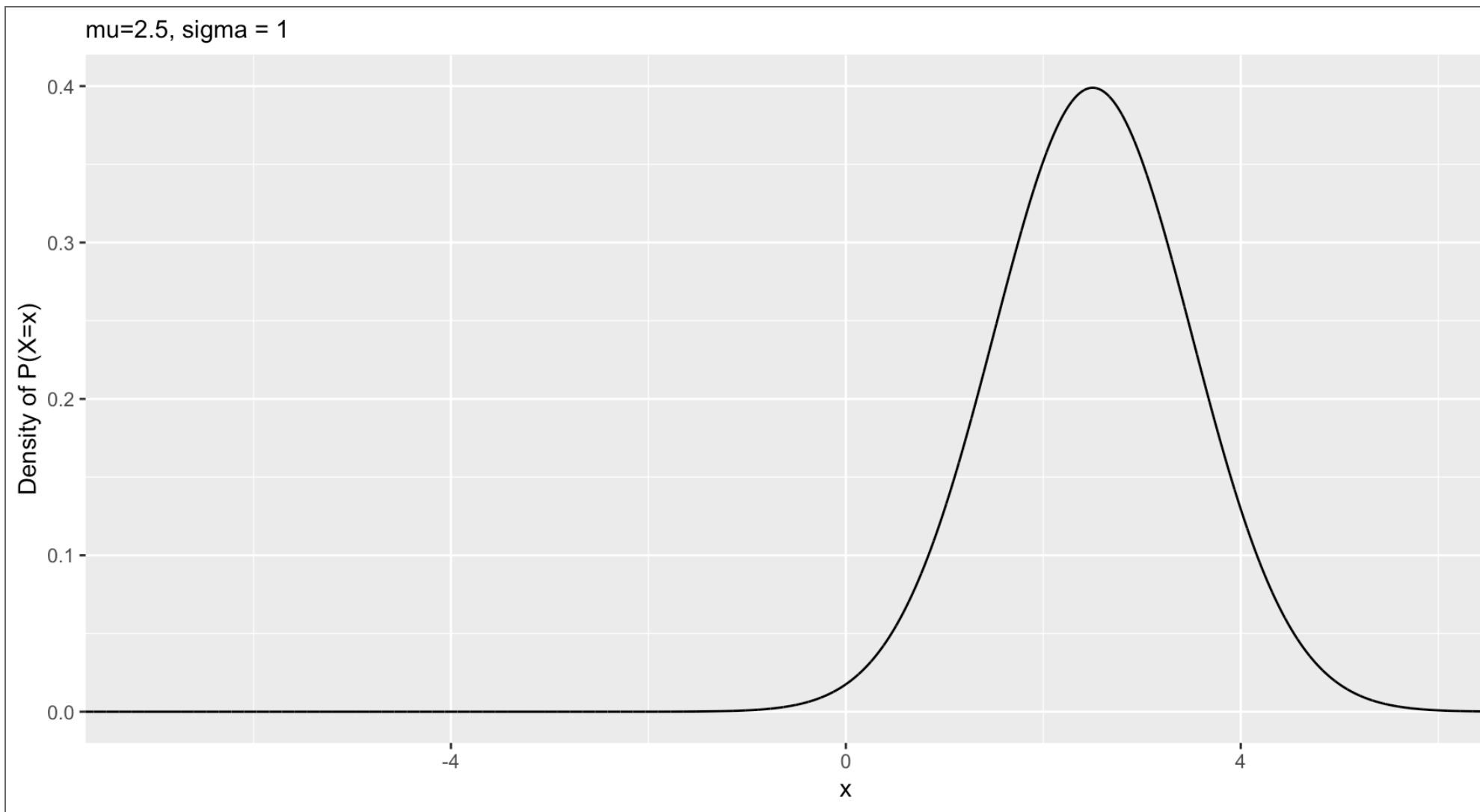
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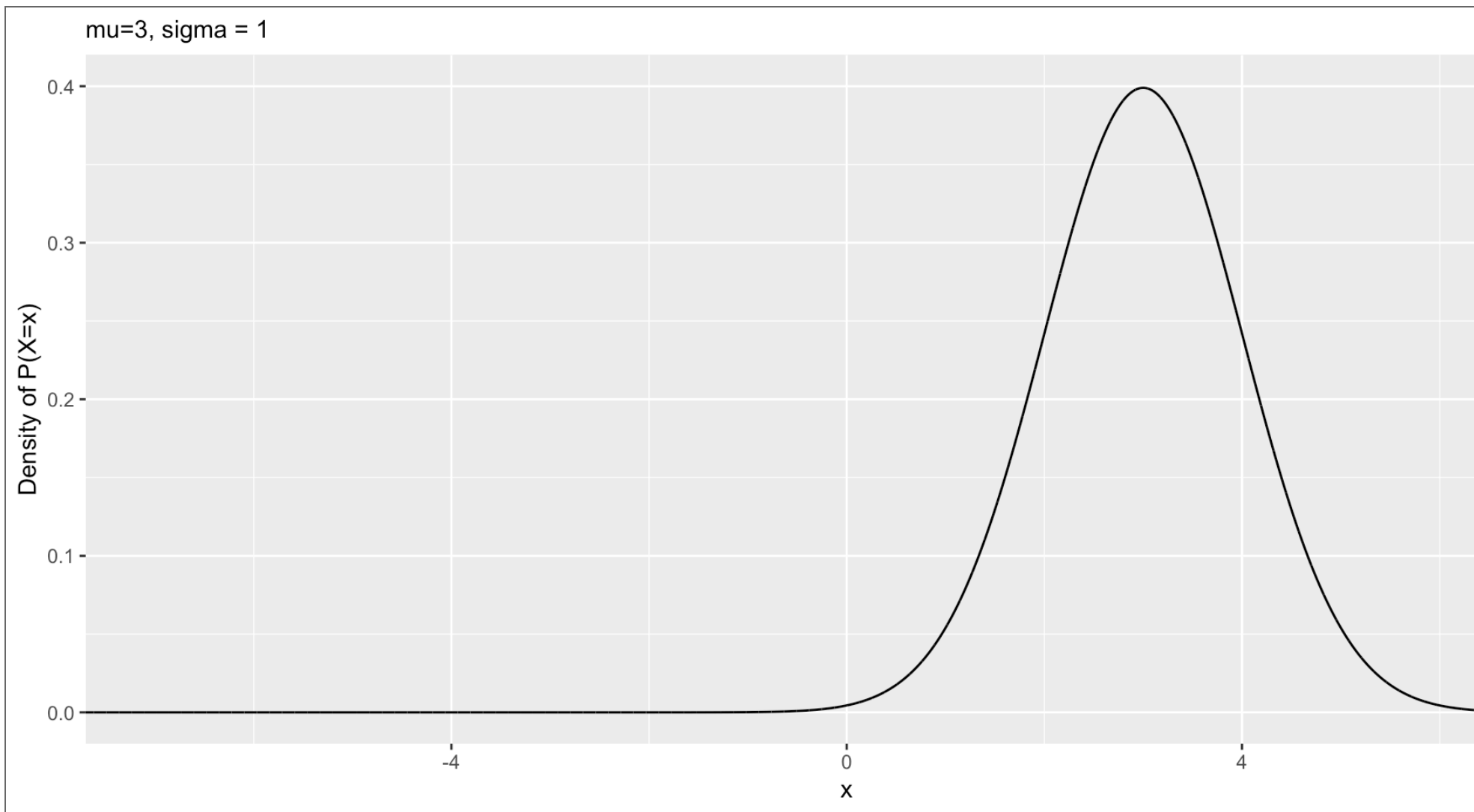
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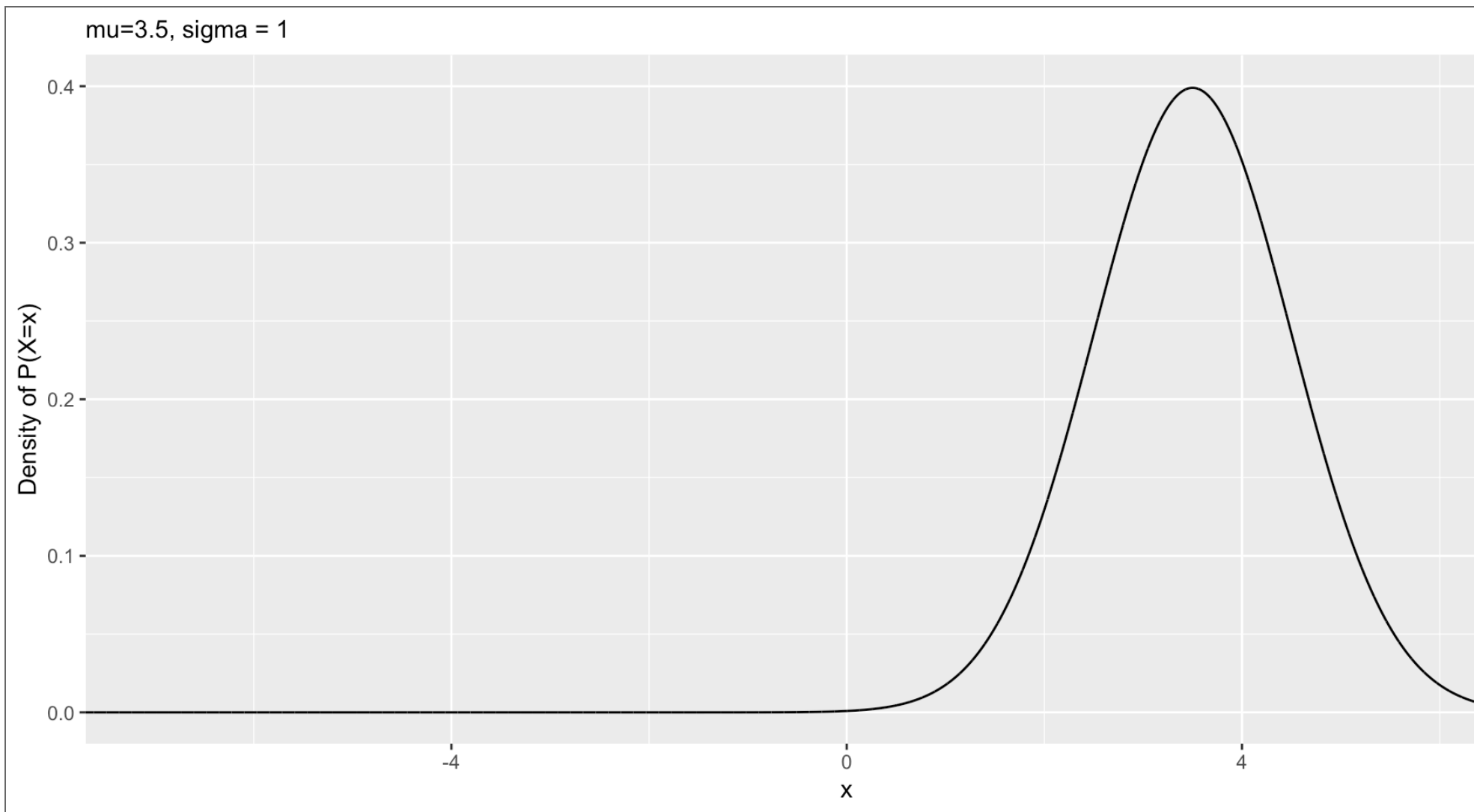
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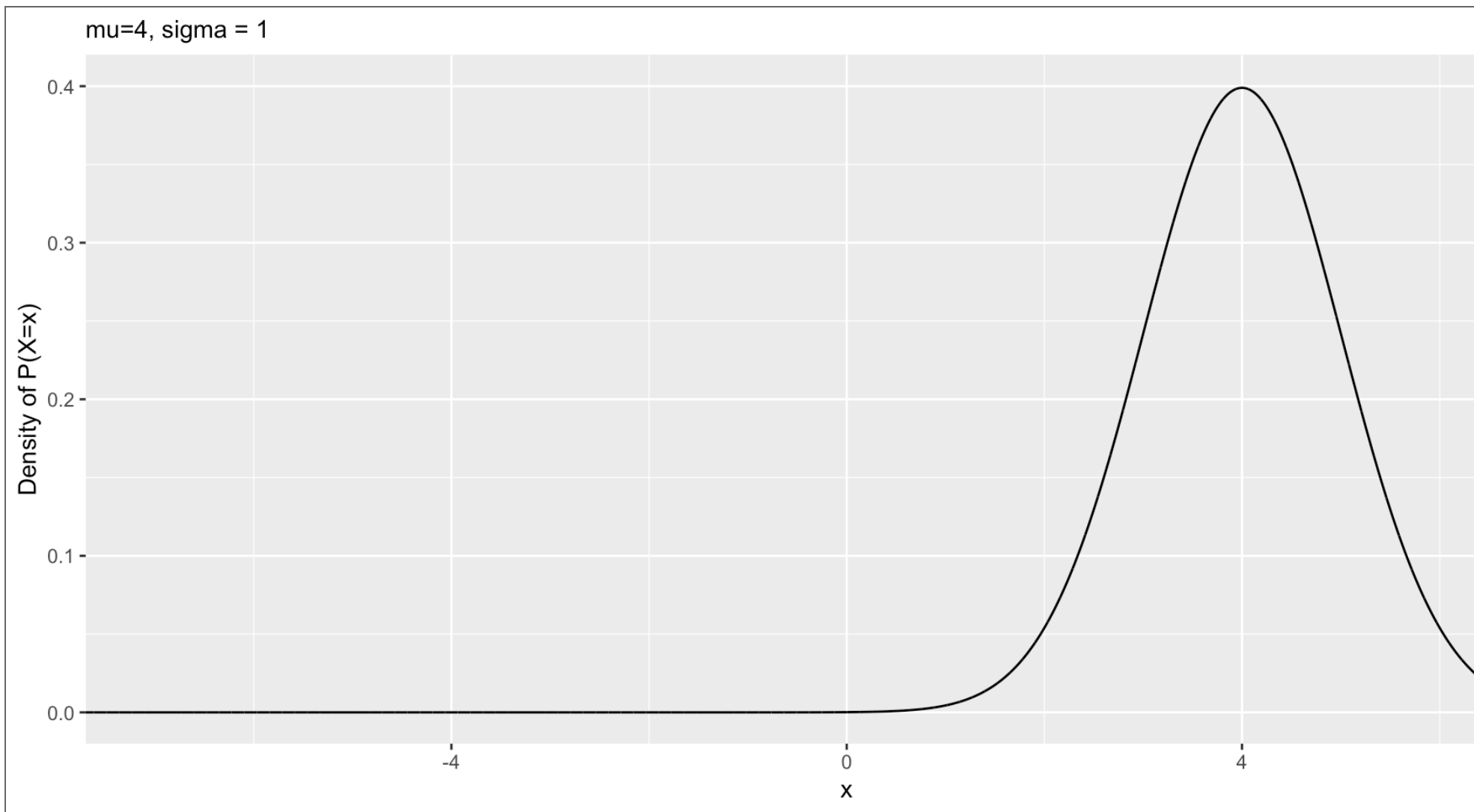
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The variance parameter

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The Normal Cumulative Distribution Function

Interpreting the PDF and CDF

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Z-scores and area under the curve

- To obtain a z-score, we subtract the mean and divide by the standard deviation:

$$\text{z-score} = \frac{X - \mu}{\sigma}$$

- For a Normal variable, z-scores are distributed $z \sim N(0, 1)$
- What does a z-score of 0 indicate? -1? 2?

Special features of Normal distributions:

- The sum of many random variables from other distributions are often Normal
- If X is a Normal random variable, then $Y = X + c$ is also Normal
 - $Y \sim (\mu + c, \sigma^2)$
- It works for multiplication too, as a scaling factor
 - $Y = cX$ is distributed $Y \sim N(c\mu, (c\sigma)^2)$
- Z-scores of a Normal random variable are $N(0, 1)$

Correlation

Correlation defined

- Correlation measures the linear relationship between two variables x and y

$$cor_{xy} = \frac{E[(x - \bar{x})(y - \bar{y})]}{\sigma_x \cdot \sigma_y}$$

Correlation and Z-scores

- It's helpful to have measures on the same scale
- A z-score measures the number of standard deviations an observation is away from the mean of a variable.
- z score of $x_i = \frac{x_i - \bar{x}}{S_x}$
- In R, we can transform a numeric into a z-score using `scale()`

Z-scores in R

```
1 iris |>  
2   mutate(Sepal.Length.sc = scale(Sepal.Length)) |>  
3   select(Sepal.Length, Sepal.Length.sc)
```

	Sepal.Length	Sepal.Length.sc
1	5.1	-0.89767388
2	4.9	-1.13920048
3	4.7	-1.38072709
4	4.6	-1.50149039
5	5.0	-1.01843718
6	5.4	-0.53538397
7	4.6	-1.50149039
8	5.0	-1.01843718
9	4.4	-1.74301699
10	4.9	-1.13920048
11	5.4	-0.53538397
12	4.8	-1.25996379
13	4.8	-1.25996379
14	4.3	-1.86378030
15	5.8	-0.05233076
16	5.7	-0.17309407
17	5.4	-0.53538397
18	5.1	-0.89767388
19	5.7	-0.17309407
20	5.1	-0.89767388
21	5.4	-0.53538397

Z-score transformed distributions have the same shape as the original data

Correlation

- Correlation measures the linear association between two variables. We often use the letter r to denote a correlation.

$$r(x, y) = \frac{1}{n} \sum_{i=1}^n \frac{x_i - \bar{x}}{S_x} \times \frac{y_i - \bar{y}}{S_y}$$
$$= E[z(x) \times z(y)]$$

In R, you can use `cor()`

Evaluate correlations using `cor()`

- Compute the correlation between `Sepal.Length` and `Sepal.Width`. What does it mean?
- Compute the correlation between `Petal.Length` and `Petal.Width`. What does it mean?
- Compute the correlation between `Petal.Length` and `Sepal.Width`. What does it mean?

Interpreting relationships with scatterplots

Correlation and scatterplots

Correlation and scatterplots

Correlation and scatterplots

Correlation and Scatterplots

Return to iris

- Scatterplot Sepal.Width and Petal.Width
- How would you describe the relationship between the variables?
- Estimate the correlation
- Describe what you find using both the estimated correlation and your interpretation of the scatterplot

Why we visualize

Exercise

- `load `data("anscombe")`
- Compute correlations for each pair $x_1, y_1; x_2, y_2; \dots$
- What do you find?
- Now scatterplot each pair using `ggplot`

Homework prep

- Download the Uniform Crime Reports Arrests by Age, Sex, and Race 2022 data from NACJD in delimited format:
<https://www.icpsr.umich.edu/web/NACJD/studies/39063>
- Move the data into a subdirectory of your intro_stats folder called ./data/

Speaker notes